Adaptive Hutch^{++}: Query Allocation, Subspace Capture, and Risk Certification

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Faculty advisor: Prof. Swati Padmanabhan

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Abstract

Adaptive query allocation in Hutch^{++} requires balancing the improvement in a randomized low-rank approximation against the loss of residual trace samples. We study this balance through exact rank-aware accounting and realized, probe-specific conditional risk. We derive a local criterion for accepting an additional basis direction, relate residual energy to dominant-subspace capture, and construct a pilot transcript that cannot distinguish a captured spectral knee from a missed leading direction. Frozen synthetic experiments show that rare range-capture failures can move finite empirical mean-risk minimizers beyond the ideal spectral knee, even when the sampled basis has full numerical rank. Direct sample-variance estimates provide useful empirical discrimination between realized Rademacher risks, but charging for this information can worsen most individual paths while improving aggregate mean risk. We prove conditional safe-acceptance statements and identify small-budget vacuity in two explicit confidence constructions. A paired common-probe formulation and a structural-prior bound clarify what remains unresolved. These results characterize limitations of adaptive allocation and certification; they do not establish a uniformly superior allocator or an impossibility theorem for all small-budget certificates.

1 Introduction

We consider trace estimation when a matrix can be accessed only through matrix–vector products. Under a fixed query budget, low-rank variance reduction creates an allocation problem: a query used to learn a basis is no longer available to sample the residual. The relevant question is not simply whether a matrix has a rapidly decaying spectrum. It is whether the realized basis reduces the residual estimator risk enough to compensate for the queries spent constructing it.

1.1 Background and problem formulation

Hutch^{++} combines a randomized low-rank trace contribution with stochastic estimation of the complementary trace. Meyer et al. [4] establish improved matrix–vector query complexity for positive semidefinite matrices. Our starting point is this decomposition, rather than a competing claim of improved worst-case query complexity.

In a basic implementation, q queries form a range sketch, r queries evaluate the accepted rank-r basis, and ℓ queries estimate the residual. Thus $q + r + \ell = m$. The familiar denominator $m - 2q$ is a full-rank special case, not the general ledger. Moreover, the equalities $r = q$ and $q = r_\star$, where $r_\star$ is a dominant spectral rank, do not imply that the randomized basis accurately captures that dominant eigenspace.

This distinction leads to one central question: under a fixed matrix–vector query budget, when is an additional subspace direction worth its construction and evaluation cost, and what can be certified about that decision? We split this question into two parts. First, what exact conditional-risk reduction justifies another low-rank direction? Second, can that reduction be certified from a small number of additional queries? The second question includes an economic constraint: confidence information uses queries, and an abstaining procedure cannot recover those queries by returning to a baseline basis.

1.2 Results and scope

We separate the contributions into a mathematical framework and empirical evidence. The mathematical framework has three parts.

1. **Exact allocation and marginal-risk identities.** We state the conditions under which random, sequentially chosen allocations remain unbiased. We distinguish Gaussian and Rademacher conditional risk and show that an accepted direction is beneficial precisely when its fractional residual-energy reduction exceeds the fractional loss of residual capacity (Proposition 3.1).

2. **Capture mechanisms beyond numerical rank.** For step spectra, a nonnegative residual-energy decomposition separates tail energy from signal leakage. A graph representation explains the effect of a poorly conditioned square signal sketch. A separate exact construction shows that a leading eigendirection can be absent from an otherwise full-rank pilot (Proposition 4.3).

3. **Certification targets and cost.** We derive the moments and Hoeffding decomposition of sample variance, formulate paired risk differences, and give safe-acceptance implications under simultaneous confidence events. We distinguish those implications from complete-policy safety after construction and certification costs (Propositions 5.3 and 5.4).

The empirical evidence then tests this framework and records its limits. Two audited confidence constructions are analytically vacuous on the small-budget grid. The experiments nevertheless find useful risk information, substantial common-probe covariance reduction, and rare-path protection. We report the accompanying failures and limitations rather than interpreting these observations as a universal guarantee.

Many underlying identities are elementary or classical. We do not claim them individually as new mathematical discoveries. Their role is to make the allocation decision, the probe-specific risk target, and the cost of certification explicit in a common framework. The contribution of the experimental analysis is a controlled separation of failure mechanisms, including hypotheses that the recorded results did not support.

1.3 Relation to prior work

Randomized range finding. Halko, Martinsson, and Tropp [3] develop a general framework for randomized low-rank approximation. We use a self-contained step-spectrum calculation to study capture, rather than importing a Gaussian sketch theorem for a rotated coordinate-Rademacher block.

Quadratic-form concentration. Cortinovis and Kressner [2] provide the quadratic-chaos tail bound used in our truncation calculation. Hypercontractivity supplies degree-dependent moment control [5]. Neither ingredient, without the additional steps analyzed below, is a useful small-budget certificate for the realized risk comparison.

Organization and boundary. We follow a theory-first exposition: preliminaries, allocation analysis, capture mechanisms, certification, and experimental validation, with supporting calculations and provenance in appendices. The detailed experimental records are unpublished project artifacts [1]. All empirical claims are conditional on their stated finite populations. We do not claim a universal one-column oversampling rule, a new minimax lower bound, or a completed confidence-certified online allocator.

2 Query accounting and the estimator

Let $A \in \mathbb{R}^{d\times d}$ be fixed and real symmetric. Positive semidefiniteness is needed for the ordered positive-eigenvalue models, but not for the trace and variance identities. Let $Q \in \mathbb{R}^{d\times r}$ have orthonormal columns. Empty Q is permitted. Define

$$R_Q = I - QQ^\top, \qquad H_Q = R_Q A R_Q. \tag{2.1}$$

These matrices are mathematical objects; production implementations apply the projectors implicitly rather than materializing dense residual matrices.

Table 1: Notation and conditioning

Symbol	Meaning and distinction
m	Total estimator query budget
S	Range-sketch matrix; uppercase S is reserved for this object
q	Attempted sketch columns / products forming AS
$r = r_q$	Accepted basis rank; $r \le q$, not automatically q
$k = r_\star$	Dominant rank in a synthetic step model, not accepted rank
B	Stopped reusable pilot width; $B \le q$
ℓ	Fresh final residual-probe count
s	Fresh certification-probe count, separate from ℓ
c_{pre}	Actual committed candidate-construction cost
$\mathcal{F}$	Information fixing candidate actions before certification
$\mathcal{G}$	Information fixing the selected action before final estimation, including certification if used
$\widehat{\sigma}^2_{x,s}$	Unbiased sample variance for action x from s certification probes

Conditioning on $\mathcal{F}$ or $\mathcal{G}$ freezes the corresponding actions and their residual denominators. Fresh probes are independent of that information and conditionally iid. A random pilot stage or allocation is allowed; freshness, rather than a deterministic stopping time, is the essential condition.

Proposition 2.1 (Exact construction accounting)**.** *Pilot products are retained inside the final q-column range sketch. An accepted r-column basis is used, and AQ is queried once and cached. No additional oracle calls are omitted from the ledger.*

The basic architecture has

Algorithm 1 Frozen-action trace estimation: accounting template

Input: symmetric matrix–vector oracle, budget m, a range policy using q products.
Output: an unbiased trace estimate, provided $\ell = m - q - r > 0$.

1. Construct AS with q oracle products, retaining any reusable pilot prefix.
2. Compute an orthonormal accepted basis Q of rank r and query/cache AQ with r products.
3. Freeze the action and set $\ell = m - q - r$.
4. Draw ℓ fresh independent isotropic probes g_j.
5. Form $h_j = g_j - Q(Q^\top g_j)$, query Ah_j, and return $\operatorname{tr}(Q^\top AQ) + \ell^{-1} \sum_j h_j^\top Ah_j$.

$$\boxed{q + r + \ell = m, \qquad \ell = m - q - r > 0.} \tag{2.2}$$

The special case $\ell = m - 2q$ holds only when $r = q$. A rejected sketch direction costs one query even when it gains no rank.

Proof. A reusable pilot of width B followed by $q - B$ new columns costs $B + (q - B) = q$, not $B + q$. Computing AQ costs r additional products, and final residual estimation costs ℓ. Adding these disjoint costs gives the identity. There is no cancellation of an attempted query merely because QR rejects its direction.

□

Retaining a pilot of width B restricts the final sketch to $q \geq B$. For nested candidates sharing all required products, $c_{\text{pre}} = \max_x(q_x + r_x)$; otherwise, c_{pre} is the actual committed query count.

Theorem 2.2 (Conditional unbiasedness, including sequential decisions)**.** *Q and the positive integer ℓ are $\mathcal{G}$-measurable. Each final probe satisfies $\mathbb{E}[g_j g_j^\top \mid \mathcal{G}] = I$. The requisite expectations exist.*

For

$$\widehat{t} = \operatorname{tr}(Q^\top AQ) + \frac{1}{\ell} \sum_{j=1}^{\ell} g_j^\top H_Q g_j, \tag{2.3}$$

we have

$$\mathbb{E}[\widehat{t} \mid \mathcal{G}] = \operatorname{tr}(A), \qquad \mathbb{E}\widehat{t} = \operatorname{tr}(A). \tag{2.4}$$

Proof. Conditional isotropy gives $\mathbb{E}[g_j^\top H_Q g_j \mid \mathcal{G}] = \operatorname{tr}(H_Q)$. Because $R_Q^2 = R_Q$, cyclicity of trace yields

$$\operatorname{tr}(H_Q) = \operatorname{tr}(AR_Q) = \operatorname{tr}(A) - \operatorname{tr}(Q^\top AQ). \tag{2.5}$$

Thus the two pieces of the estimator add in conditional expectation to $\operatorname{tr}(A)$. Taking another expectation proves the unconditional assertion.

□

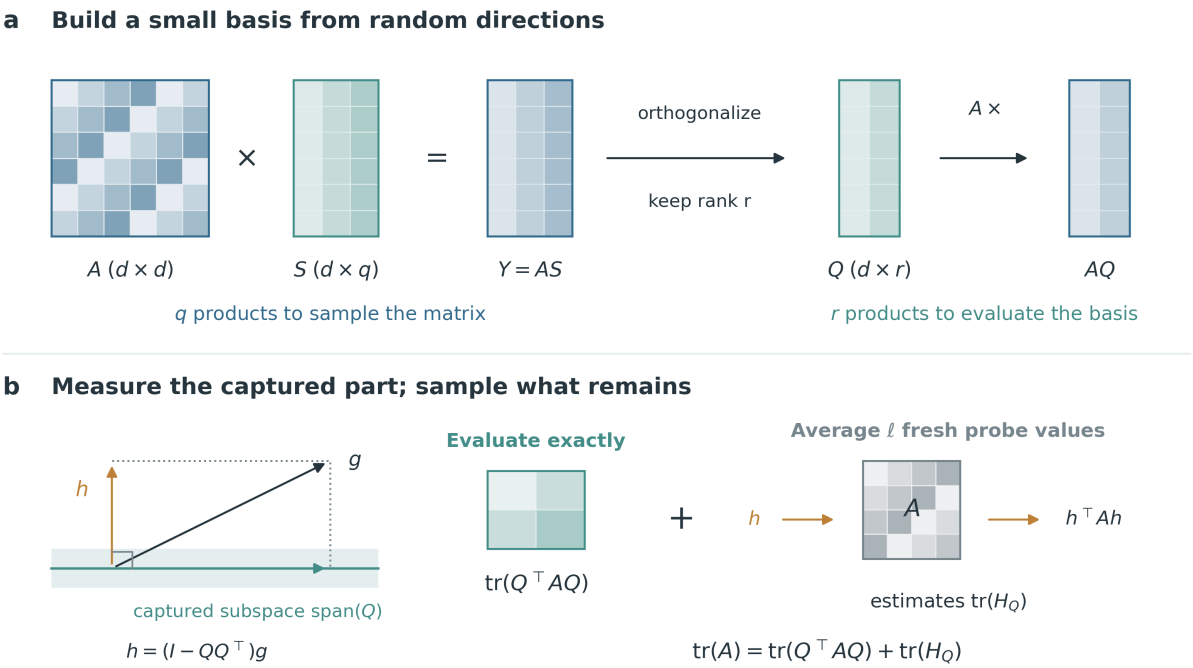


Figure 1: **What each query buys.** (a) A random sketch constructs a basis with accepted rank $r \leq q$; evaluating AQ costs another r queries. (b) The captured trace contribution is evaluated exactly, while fresh projected probes estimate the complementary trace. The geometry is schematic and does not assert exact dominant-eigenspace capture. (c) Certification consumes s additional queries, leaving $\ell_{\text{paid}} = m - c_{\text{pre}} - s$. Bar widths are schematic. Returning to a baseline basis does not refund committed queries.

Interpretation. The decision may use a stopped pilot or a certification batch. It must be completed before drawing the final probes. Reusing the selection probes as final probes invalidates this proof unless an additional argument restores conditional isotropy. Independence between final probes is needed for the variance formula below, but not for the expectation calculation itself. No optional-stopping theorem is invoked.

Theorem 2.3 (Exact conditional Gaussian and Rademacher risk)**.** *Those of Theorem 2.2, with conditionally independent final probes. They are either standard Gaussian or independent coordinate Rademacher signs. Define*

$$E_G(Q) = \|H_Q\|_F^2, \qquad E_R(Q) = \sum_{i \neq j} (H_Q)_{ij}^2. \tag{2.6}$$

The conditional mean-squared errors are exactly

$$\boxed{\mathcal{R}_G(Q) = \frac{2E_G(Q)}{\ell}, \qquad \mathcal{R}_R(Q) = \frac{2E_R(Q)}{\ell}.} \tag{2.7}$$

Their expectations over basis and decision randomness equal unconditional MSE.

Proof. For a Rademacher vector,

$$g^\top H_Q g - \operatorname{tr}(H_Q) = 2 \sum_{i<j} (H_Q)_{ij} g_i g_j. \tag{2.8}$$

For distinct unordered pairs, the corresponding products have zero cross-expectation: some independent sign occurs an odd number of times. Each individual product has second moment one. Hence the single-probe variance is $4\sum_{i<j}(H_Q)_{ij}^2 = 2E_R(Q)$. For Gaussian probes, orthogonally diagonalize H_Q. Rotational invariance reduces the centered quadratic form to $\sum_i \lambda_i(z_i^2 - 1)$, whose independent terms have variances $2\lambda_i^2$. Its variance is therefore $2E_G(Q)$. Averaging ℓ independent terms divides either variance by ℓ. Theorem 2.2 makes conditional bias zero. The law of total expectation applied to squared error then gives unconditional MSE.

□

Interpretation and boundary cases. Gaussian/Rademacher here describes the **final probes**, not the range sketch. For the same Q and ℓ, $E_R \leq E_G$; different allocations or different bases cannot be compared by that inequality alone. A diagonal H_Q has zero Rademacher variance even if it has positive Gaussian variance. A diagonal A need not leave a diagonal H_Q after randomized projection. These facts will matter in Section 7.

3 Rank-aware allocation analysis

For PSD A, order its eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ and write $T(j) = \sum_{i>j} \lambda_i^2$. On a common allocation grid with $0 \leq r_q \leq q \leq d$ and strictly positive denominators, compare

$$\mathcal{R}_{\text{full}}(q) = \frac{2T(q)}{m - 2q}, \qquad \mathcal{R}_{\text{rank}}(q) = \frac{2T(r_q)}{m - q - r_q}, \tag{3.1}$$

$$\mathcal{R}_G(q) = \frac{2E_G(Q_q)}{m - q - r_q}, \qquad \mathcal{R}_R(q) = \frac{2E_R(Q_q)}{m - q - r_q}. \tag{3.2}$$

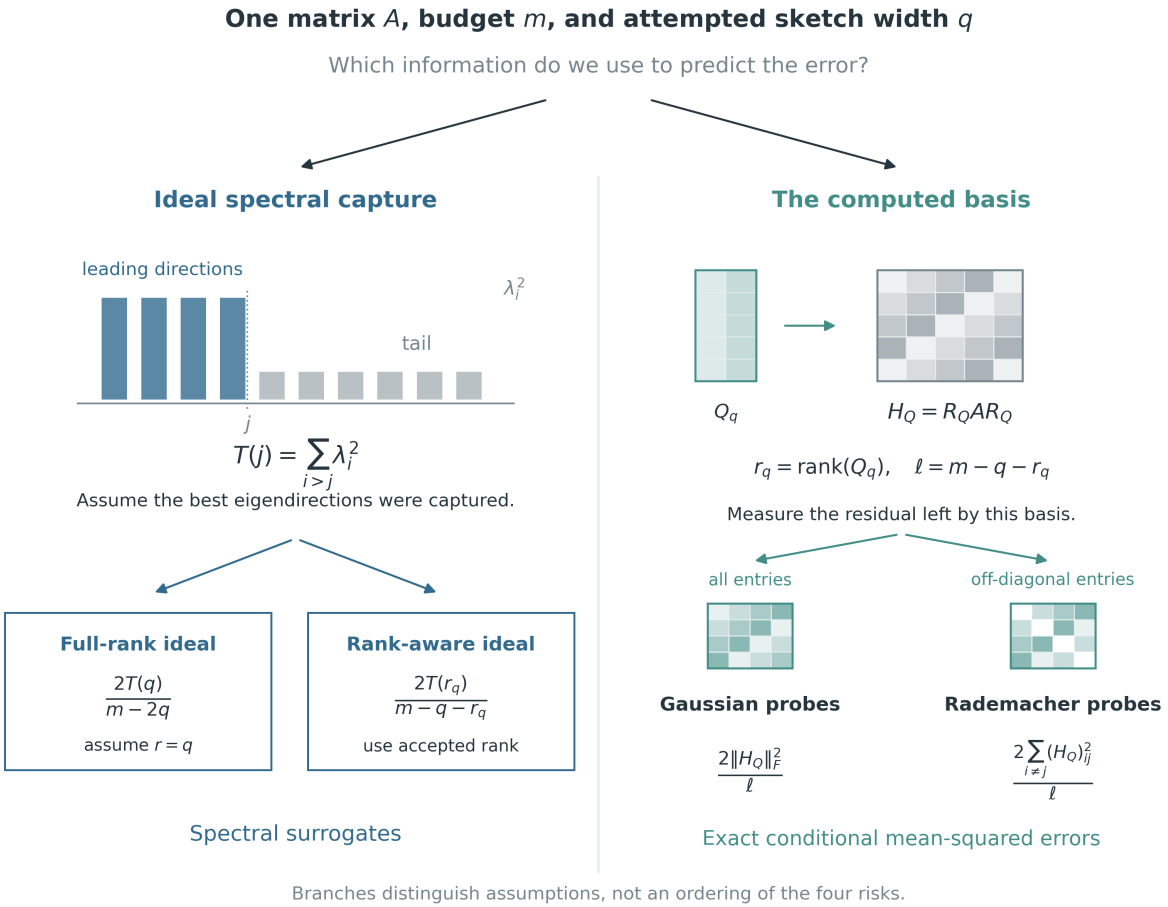


Figure 2: **Four allocation objectives.** Spectral models assume ideal leading-eigenspace capture; realized models use the computed basis Q_q. The right-hand masks highlight variance contributions: all entries of H_Q for Gaussian probes, and off-diagonal entries for Rademacher probes. Diagonal entries still contribute to the trace in both cases. The spectrum is schematic. All denominators are positive; branches distinguish assumptions, not an ordering of the four risks.

The first imagines one perfectly targeted direction per sketch query. The second corrects accepted rank but still assumes ideal targeting. The last two use the actual basis and are exact conditional risks for their respective final-probe laws.

Proposition 3.1 (Realized successful and failed marginal identities)**.** *Put $D = m - q - r$. A successful extension adds one attempted sketch and one accepted direction, yielding $(q+1, r+1, Q')$, and $D > 2$. Use either $E = E_G$ or $E = E_R$ consistently.*

Writing $E = E(Q)$ and $E' = E(Q')$,

$$\mathcal{R}' - \mathcal{R} = -\frac{2\{D(E - E') - 2E\}}{D(D-2)}. \tag{3.3}$$

If $E > 0$, the extension is beneficial exactly when

$$\boxed{\frac{E - E'}{E} > \frac{2}{D}} \quad \Longleftrightarrow \quad \boxed{\frac{E'}{E} < \frac{D-2}{D}.} \tag{3.4}$$

For a failed-rank extension with ***unchanged basis projector*** *and* $D > 1$*,*

$$\mathcal{R}' - \mathcal{R} = \frac{2E}{D(D-1)} \geq 0. \tag{3.5}$$

Proof. For success, subtract $2E/D$ from $2E'/(D-2)$ and collect the numerator. Its denominator is positive, so the sign is the opposite of $D(E - E') - 2E$. Division by DE is permissible when $E > 0$ and gives the equivalent inequalities. For failure, the unchanged projector keeps the numerator fixed; subtracting $2E/D$ from $2E/(D-1)$ gives the stated expression.

□

Interpretation. A successful direction costs two final samples: one sketch query and one AQ query. Its fractional energy reduction must exceed $2/D$. A rejected direction costs one sample without changing the residual. If $E = 0$, no nonnegative risk can strictly improve it; a failed direction remains at zero rather than being strictly harmful. These cases must not be hidden by adding a numerical floor.

For $m = 160, q = r = k = 15$, $D = 130$. A successful extra direction pays for itself if $E'/E < 128/130 \approx 0.984615$, or the energy drops by more than 1.53846%. The identity applies to each coupled path. A condition using ratios of empirical mean energies addresses a mean-risk comparison, not the fraction of individually improved paths.

A useful caution. Nested subspaces need not make Rademacher residual energy decrease. Let $A = I_2$, initially Q empty. Then $E_R = 0$. Adding $u = (1,1)^\top/\sqrt{2}$ leaves $H = I - uu^\top$ with two off-diagonal entries $-1/2$, so $E'_R = 1/2$. Projection reduced total energy but created off-diagonal energy. Thus Proposition 3.1 does not assume an energy decrease; it tests whether one occurred and was large enough.

Corollary 3.2 (Ideal spectral marginal and step knee)**.** *Consider the ideal Gaussian/Frobenius risk model, in which accepted directions are leading eigenvectors. A successful extension captures* λ_{r+1}*. Then* $E = T(r)$ *and* $E' = T(r) - \lambda_{r+1}^2$*.*

The ideal successful extension is beneficial if and only if

$$(m - q - r)\lambda_{r+1}^2 > 2T(r). \tag{3.6}$$

In the full-rank ideal model the marginal numerator is nonincreasing on the feasible adjacent grid. For the step spectrum with $0 < \eta < 1$*, the condition* $2k + 2(d-k)\eta^2 < m < 2d$ *makes the feasible knee* k *the unique minimizer.*

Proof. Substitution into Proposition 3.1 gives

$$\boxed{(m - q - r)\lambda_{r+1}^2 > 2T(r).} \tag{3.7}$$

Only when $r = q$ does this reduce to $(m-2q)\lambda_{q+1}^2 > 2T(q)$. Define $M(q) = (m-2q)\lambda_{q+1}^2 - 2T(q)$. Direct subtraction gives

$$M(q+1) - M(q) = (m - 2q - 2)(\lambda_{q+2}^2 - \lambda_{q+1}^2) \leq 0 \tag{3.8}$$

on the feasible adjacent full-rank grid. Hence this ideal objective has a single-crossing marginal sign, with ties possible.

For the step spectrum $\lambda_{1:k} = 1$, $\lambda_{k+1:d} = \eta$, with $0 < \eta < 1$, the pre-knee marginal numerator is $m - 2k - 2(d-k)\eta^2$, and the post-knee numerator is $\eta^2(m-2d)$. If

$$2k + 2(d-k)\eta^2 < m < 2d \tag{3.9}$$

and the relevant allocations are feasible, risk decreases up to k and increases after it. The unique ideal full-rank minimizer is k. At $\eta = 0$, the post-knee risk can instead be identically zero; uniqueness must not be asserted.

□

The corollary concerns the ideal tail model, not the capture quality of a randomized basis. It also does not identify the Rademacher residual energy with $T(r)$. For example, $A = \mathrm{diag}(2, 1)$ and $Q = e_1$ give $T(1) = 1$ but $E_R(Q) = 0$.

4 Subspace capture beyond numerical rank

Lemma 4.1 (Structured residual energy)**.** *Let $A = \eta I + (1-\eta)U_1U_1^\top$, where U_1 has k orthonormal columns and $0 \le \eta < 1$. Complete it to an orthogonal basis $[U_1, U_2]$. Define $Z_Q = R_Q U_1$.*

For any orthonormal Q,

$$H_Q = \eta R_Q + (1-\eta) Z_Q Z_Q^\top, \tag{4.1}$$

$$\boxed{E_G(Q) = \eta^2(d-r) + 2\eta(1-\eta)\|Z_Q\|_F^2 + (1-\eta)^2\|Z_Q^\top Z_Q\|_F^2.} \tag{4.2}$$

Proof. Insert the structured A into R_QAR_Q. In the squared Frobenius norm, $\|R_Q\|_F^2 = d - r$, $R_QZ_Q = Z_Q$, and $\|Z_QZ_Q^\top\|_F^2 = \|Z_Q^\top Z_Q\|_F^2$. Expanding yields the formula. Each term is nonnegative.

□

Why this matters. The formula exhibits leakage directly and avoids subtracting several order-one numbers to recover a tiny residual energy. Tiny positive tails are not canonicalized to exact zero. The $\eta = 0$ zero-risk treatment in the frozen implementation also requires its stated capture/backward-residual checks.

Lemma 4.2 (Square-sketch graph and principal angles)**.** *Let $A = \eta I + (1-\eta)U_1U_1^\top$, where U_1 has k orthonormal columns and $0 \le \eta < 1$. Complete it to an orthogonal basis $[U_1, U_2]$. Define $Z_Q = R_QU_1$.*

Let $S_1 = U_1^\top S$, $S_2 = U_2^\top S$. If $q = k$ and S_1 is invertible, set $F = \eta S_2 S_1^{-1}$. For Q spanning AS,

$$\boxed{\|R_QU_1\|_2 = \frac{\|F\|_2}{\sqrt{1 + \|F\|_2^2}}.} \tag{4.3}$$

Proof. In the eigenbasis, AS is represented by $[S_1^\top, \eta S_2^\top]^\top$. Multiplication on the right by S_1^{-1} does not change its range, which is therefore the graph of F, spanned by $[I, F^\top]^\top$. An orthonormal basis is

$$\begin{bmatrix} I \\ F \end{bmatrix} (I + F^\top F)^{-1/2}. \tag{4.4}$$

Its overlap with the signal coordinate space has singular values $(1+\sigma_i(F)^2)^{-1/2}$. These are the cosines of the principal angles, so the largest sine is exactly the displayed expression.

□

Full rank of S_1 does not imply a small inverse. Poor conditioning can amplify a nonzero tail and create leakage, which Lemma 4.1 converts to residual energy. The product $S_2 S_1^{-1}$, not S_1^{-1} alone, determines the graph error.

For $q > k$ and full row rank S_1, the range contains the graph with factor $\eta S_2 S_1^\dagger$. It may contain additional useful directions, so the square-case equality becomes a capture bound, not generally an equality for the whole range. On a nested path, appending a signal column v changes the Gram matrix to $S_1 S_1^\top + vv^\top$. The minimum eigenvalue cannot decrease; thus the full-row-rank pseudoinverse norm cannot increase. Neither strict improvement nor good conditioning is guaranteed. Coordinate-Rademacher S also does not make the rotated block $U_1^\top S$ iid Gaussian or iid Rademacher.

Proposition 4.3 (A missing signal direction can be invisible to the pilot)**.** *A is symmetric, w is a unit vector with $Aw = w$, and $S^\top w = 0$. Let Q span AS. For the PSD step model, w is one of its unit-eigenvalue directions and $0 \le \eta < 1$.*

In exact arithmetic,

$$Q^\top w = 0, \quad R_Q w = w, \quad H_Q w = w. \tag{4.5}$$

Moreover, $A' = A - (1-\eta) ww^\top$ replaces this eigenvalue by η and has the same observed products:

$$A'S = AS, \qquad A'Q = AQ. \tag{4.6}$$

Proof. Symmetry gives $(AS)^\top w = S^\top A w = S^\top w = 0$. Thus w is orthogonal to the sampled range and to Q. The first three identities follow by projecting $Aw = w$. In the two product differences, $ww^\top S = 0$ and $ww^\top Q = 0$, proving transcript equality. In the step model, modifying one eigenvalue in its own orthogonal eigendirection leaves the other eigenvalues unchanged and preserves PSD.

□

A deterministic pilot decision based only on those products cannot distinguish the two matrices on this transcript. This is a local identifiability limitation: the alternative can depend on the realized sketch. It does not exclude probabilistic guarantees or additional fresh queries. The positive-oversampling example in Section 7.2 differs from the full-row-rank conditioning mechanism of Lemma 4.2.

5 Realized-risk estimation and conditional safety

Condition on $\mathcal{F}$, fixing candidate a and baseline 0. For fresh Rademacher certification probes, put

$$X_{x,j} = g_j^\top H_x g_j, \qquad \sigma_x^2 = \mathrm{Var}(X_{x,j} \mid \mathcal{F}) = 2E_R(Q_x), \quad x \in \{a, 0\}. \tag{5.1}$$

Algorithm 2 Common-probe risk diagnostic for preconstructed actions

Input: cached Q_x, AQ_x for every action x, fixed positive ℓ_x, and integer $s \geq 2$.
Output: sample-variance risk estimates; not a confidence certificate.

1. Draw s fresh coordinate-Rademacher probes g_j, independent of the fixed actions.
2. Query Ag_j once per probe; share each result across all actions.
3. For each x, j, compute $h_{x,j} = g_j - Q_x Q_x^\top g_j$ and $X_{x,j} = h_{x,j}^\top [Ag_j - AQ_x(Q_x^\top g_j)]$.
4. Return $\widehat{\sigma}_{x,s}^2/\ell_x$, where $\widehat{\sigma}_{x,s}^2 = (s-1)^{-1}\sum_j (X_{x,j} - \bar{X}_x)^2$.

Cost: exactly s new oracle queries; cached products were paid during construction. Certification probes are not reused as final residual probes.

The target is $\mathcal{R}_x = \sigma_x^2/\ell_x$. Estimating a fitted decay exponent is no longer necessary: one can estimate the variance of the actual residual quadratic form.

With cached AQ_x, one product Ag serves every action:

$$h_x = g - Q_x(Q_x^\top g), \qquad Ah_x = Ag - AQ_x(Q_x^\top g), \qquad X_x = h_x^\top Ah_x. \tag{5.2}$$

The cache must exist before certification and be charged to construction. A batch of 32 products Ag_j costs 32 certification queries, regardless of action count. Certification observations are not reused as final estimator probes.

Lemma 5.1 (Exact sample-variance moments and Hoeffding decomposition)**.** *Conditional on the frozen action, $X_1, \ldots, X_s$ are iid, $s \geq 2$, with mean μ, variance σ^2, and finite fourth central moment μ_4. Define*

$$\widehat{\sigma}_s^2 = \frac{1}{s-1}\sum_{i=1}^{s}(X_i - \bar{X})^2. \tag{5.3}$$

Its expectation and variance are

$$\mathbb{E}\widehat{\sigma}_s^2 = \sigma^2, \qquad \mathrm{Var}(\widehat{\sigma}_s^2) = \frac{1}{s}\left[\mu_4 - \frac{s-3}{s-1}\sigma^4\right]. \tag{5.4}$$

Here and below moments are conditional when the action is random. Its U-statistic representation and decomposition are

$$\widehat{\sigma}_s^2 = \binom{s}{2}^{-1}\sum_{i<j}\frac{(X_i - X_j)^2}{2} = \frac{1}{s(s-1)}\sum_{i<j}(X_i - X_j)^2, \tag{5.5}$$

$$h_1(x) = \frac{(x-\mu)^2 - \sigma^2}{2}, \qquad h_2(x,y) = -(x-\mu)(y-\mu), \tag{5.6}$$

$$\widehat{\sigma}_s^2 - \sigma^2 = \frac{2}{s}\sum_i h_1(X_i) + \binom{s}{2}^{-1}\sum_{i<j} h_2(X_i, X_j). \tag{5.7}$$

Proof. Set $Z_i = X_i - \mu$, $M = \sum_i Z_i^2$, and $L = \sum_i Z_i$. Then $\widehat{\sigma}_s^2 = (M - L^2/s)/(s-1)$. Independence and centering give

$$\mathbb{E}M = s\sigma^2, \quad \mathbb{E}L^2 = s\sigma^2, \tag{5.8}$$

$$\mathbb{E}M^2 = s\mu_4 + s(s-1)\sigma^4, \tag{5.9}$$

$$\mathbb{E}L^4 = s\mu_4 + 3s(s-1)\sigma^4, \qquad \mathbb{E}(ML^2) = \mathbb{E}M^2. \tag{5.10}$$

For the last identity, every cross term from $L^2 - M$ contains at least one independent centered factor to its first power. Substitution proves unbiasedness and

$$\mathbb{E}(\widehat{\sigma}_s^2)^2 = \frac{\mu_4}{s} + \frac{s^2 - 2s + 3}{s(s-1)}\sigma^4. \tag{5.11}$$

Subtract σ^4 to obtain the variance formula.

Hoeffding decomposition. The deterministic identity $\sum_{i<j}(X_i - X_j)^2 = s\sum_i (X_i - \bar{X})^2$ gives the U-statistic formula. For $h(x,y) = (x-y)^2/2$, computing $\mathbb{E}[h(x, X')] - \sigma^2$ gives h_1. Subtracting $\sigma^2 + h_1(x) + h_1(y)$ from $h(x,y)$ gives h_2. Summing over pairs yields the displayed coefficients. Finally $\mathbb{E}[h_2(X, X') \mid X] = 0$ by centering, so only the second component is canonical/degenerate in general.

□

Interpretation. The linear projection is generally nonzero, although special distributions can make it vanish. A theorem for completely degenerate U-statistics cannot bound the full sample variance without also controlling this projection. If $\sigma^2 = 0$, then X is constant almost surely and the sample variance is exactly zero; no division by σ^2 is needed.

Proposition 5.2 (Direct common-probe risk differences)**.** *Candidate and baseline are frozen before the common iid certification probes, and their denominators are fixed and positive.*

The statistic

$$\widehat{\Delta} = \frac{\widehat{\sigma}_{a,s}^2}{\ell_a} - \frac{\widehat{\sigma}_{0,s}^2}{\ell_0} \tag{5.12}$$

is unbiased for $\Delta = \sigma_a^2/\ell_a - \sigma_0^2/\ell_0$, and is itself a paired order-two U-statistic. For $Y_x = \widehat{\sigma}_{x,s}^2/\ell_x$,

$$\mathrm{Var}(Y_a - Y_0) = \mathrm{Var}(Y_a) + \mathrm{Var}(Y_0) - 2\,\mathrm{Cov}(Y_a, Y_0). \tag{5.13}$$

Proof. Apply Lemma 5.1 separately to each action and use linearity of expectation; independence between actions is unnecessary. Subtract their U-statistic kernels to obtain the paired representation. Expansion of the centered square gives the variance identity.

□

Interpretation. Pairing reduces variance relative to independent observations with the same marginals exactly when the covariance is positive. It is not guaranteed merely by sharing probes. The mathematical independent benchmark is the sum of the two marginal variances; empirical estimates of that sum remain estimates. A cyclic shift of the same repetitions is a shifted/decorrelated comparator, not an independent sample.

Proposition 5.3 (Simultaneous bounds imply safe acceptance)**.** *On an event E of probability at least $1-\delta_{\text{joint}}$, every action in a fixed finite pre-certification candidate set satisfies $L_x \le \mathcal{R}_x \le U_x$. The baseline belongs to the set. Selection is measurable.*

Any accepted action satisfying $U_a \le L_0$ has $\mathcal{R}_a \le \mathcal{R}_0$ on E. If no candidate passes and the baseline is still feasible with the same risk used in the bounds, choosing it preserves this comparison.

Proof. On the single simultaneous event, the selected action obeys

$$\mathcal{R}_a \le U_a \le L_0 \le \mathcal{R}_0. \tag{5.14}$$

If the action is the unchanged feasible baseline, the inequality holds with equality.

□

Why simultaneity matters. The action is chosen using the estimates. Separate pointwise coverage statements without a joint event do not justify substituting a selected index. A union bound over candidates or preregistered stages supplies a valid joint event if the allocated failure probabilities sum to δ_{joint}. That also handles a random selected stage; it does not require an optional-stopping assertion.

Directed variance form. Suppose the available event states

$$\sigma_a^2 \le \frac{\widehat{\sigma}_a^2}{1-\varepsilon_{a,-}}, \qquad \sigma_0^2 \ge \frac{\widehat{\sigma}_0^2}{1+\varepsilon_{0,+}}, \tag{5.15}$$

where $0 \le \varepsilon_{a,-} < 1$ and $0 \le \varepsilon_{0,+} < \infty$. An accepted candidate is safe relative to the original baseline if

$$\boxed{\widehat{\sigma}_a^2 \le \frac{1-\varepsilon_{a,-}}{1+\varepsilon_{0,+}} \frac{\ell_{\text{paid}}}{\ell_0} \widehat{\sigma}_0^2.} \tag{5.16}$$

This follows by dividing the two valid bounds by their positive denominators and chaining inequalities. Only the candidate-underestimation radius must be below one algebraically. A very large baseline-overestimation radius is weak but not undefined.

The confidence statement bounds Pr(accept and harmful) by δ_{joint}. It does **not** automatically bound Pr(harmful | accept) by the same number. Conditioning on a possibly rare acceptance event changes the probability statement. Nor does accepted-action safety prove whole-policy safety when abstention has a cost.

Proposition 5.4 (Paid fallback is not the original baseline)**.** *Original baseline construction costs c_0, whereas candidate-first construction costs c_{pre} and certification costs s. Both residual budgets are positive:*

$$\ell_0 = m - c_0, \qquad \ell_{\text{paid}} = m - c_{\text{pre}} - s. \tag{5.17}$$

Returning to exactly the baseline basis after paying extra costs changes its risk from σ_0^2/ℓ_0 to $\sigma_0^2/\ell_{\text{paid}}$. For $\sigma_0^2 > 0$, the ratio is

$$\boxed{\frac{\mathcal{R}_{\text{fallback}}}{\mathcal{R}_{\text{original}}} = \frac{\ell_0}{\ell_{\text{paid}}}.} \tag{5.18}$$

A candidate with positive baseline variance improves net risk exactly when

$$\frac{\sigma_a^2}{\sigma_0^2} < \frac{\ell_{\text{paid}}}{\ell_0}. \tag{5.19}$$

Proof. Theorem 2.3 applies after all sunk decisions and costs are fixed. The returned basis has the same numerator, but fewer final probes whenever $c_{\text{pre}} + s > c_0$. Taking the ratio proves the first identity; cross-multiplying positive denominators proves the second.

□

If baseline risk is zero, ratios are undefined and no candidate strictly improves it; compare risks directly. Even selecting the smaller numerator cannot refund construction or certification. Useful risk information therefore need not yield a beneficial complete policy.

6 Confidence bounds and their budget limits

For a multilinear polynomial of degree at most h in independent uniform signs, hypercontractivity gives $\|P\|_4 \le 3^{h/2}\|P\|_2$. Taking $h = 2$ yields the retained quadratic-chaos fourth-moment bound $\mu_4 \le 81\sigma^4$. Taking $h = 4$ instead gives the fourth-moment factor $9^4 = 6561$. These degree-dependent constants cannot be interchanged. The imported inequality is stated in O'Donnell's hypercontractivity lecture, Corollary 1.3 [5]; the substitutions and applications below are the report's calculations.

Proposition 6.1 (The frozen 81-plus-Chebyshev certificate is budget-vacuous)**.** *Use Lemma 5.1 with $\mu_4 \le 81\sigma^4$, and require simultaneous two-action control with total failure probability δ_{joint}. Allocate $\delta_{\text{joint}}/2$ to each action. Handle zero variance directly as in Lemma 5.1.*

A valid common relative-deviation radius is

$$\varepsilon_{\text{Ch}}(s, \delta_{\text{joint}}) = \sqrt{\frac{2[80 + 2/(s-1)]}{s\delta_{\text{joint}}}}. \tag{6.1}$$

At $\delta_{\text{joint}} = 0.05$, it exceeds one for every $2 \le s \le 32$.

Proof. Lemma 5.1 gives

$$\frac{\operatorname{Var}(\widehat{\sigma}_s^2)}{\sigma^4} \le \frac{80 + 2/(s-1)}{s}. \tag{6.2}$$

Chebyshev at relative deviation ε bounds the one-action failure probability by this quantity divided by ε^2. Equating it to $\delta_{\text{joint}}/2$ yields the radius; the union bound supplies simultaneity without requiring action independence. Both $80/s$ and $2/[s(s-1)]$ decrease for $s \ge 2$. Thus the smallest radius on the grid is at 32, where

$$\varepsilon_{\text{Ch}}(32, 0.05) = \sqrt{\frac{2(80 + 2/31)}{1.6}} \approx 10.00403 > 1. \tag{6.3}$$

All smaller sample sizes are also vacuous for a positive denominator $1 - \varepsilon$.

□

The event is valid, but division by $1-\varepsilon$ cannot provide a finite multiplicative variance upper bound. This failure concerns the specified construction, not every possible certificate. Throughout, δ_{joint} denotes total simultaneous failure probability rather than a per-action budget.

6.1 Truncation of the nondegenerate projection

Assumptions and imported theorem. Let $C = H - \operatorname{diag}(H)$ be nonzero, symmetric, and zero diagonal. Then $Z = g^\top C g$, $\sigma^2 = 2\|C\|_F^2$, and $\kappa = \|C\|_2/\|C\|_F \le 1$. Cortinovis–Kressner, Theorem 2, equation (8) [2] bounds its tail by

$$\Pr(|Z| \ge t) \le 2\exp\left[-\frac{t^2}{8\|C\|_F^2 + 8t\|C\|_2}\right]. \tag{6.4}$$

Substitute $t = u\sigma$ and define $V = Z^2/\sigma^2$. Then $\mathbb{E}V = 1$ and

$$\Pr(V \ge v) \le p_\kappa(v) = \min\left\{1, 2\exp\left[-\frac{v}{4 + 4\sqrt{2v}\,\kappa}\right]\right\}. \tag{6.5}$$

The zero-variance branch is deterministic and needs no normalization. The normalized nondegenerate Hoeffding component is exactly $s^{-1}\sum_i (V_i - 1)$, since $2h_1/\sigma^2 = V - 1$.

Truncation calculation. For $T > 0$, define $V^{(T)} = \min(V, T)$ and

$$a_T = (4\sqrt{2}\kappa + 4/\sqrt{T})^{-1}, \qquad \beta_\kappa(T) = 4e^{-a_T\sqrt{T}}\left(\frac{\sqrt{T}}{a_T} + \frac{1}{a_T^2}\right). \tag{6.6}$$

For $v \ge T$, the tail exponent is at least $a_T\sqrt{v}$. Therefore

$$\mathbb{E}(V - V^{(T)}) = \int_T^\infty \Pr(V > v)\,dv \le \int_T^\infty 2e^{-a_T\sqrt{v}}\,dv = \beta_\kappa(T). \tag{6.7}$$

The last equality follows by substituting $z = \sqrt{v}$ and integrating $4ze^{-a_T z}$. Hence $\mathbb{E}V^{(T)} \ge \max(0, 1 - \beta_\kappa(T))$. The retained fourth-moment and bounded-range controls give the capped variance envelope

$$\nu_\kappa(T) = \min\left\{81, \frac{T^2}{4}, 81 - \max(0, 1 - \beta_\kappa(T))^2\right\}. \tag{6.8}$$

The third term subtracts a lower bound on the squared mean from an upper bound on the second moment. For the lower-tail Bernstein bound, $\mathbb{E}V^{(T)} - V^{(T)} \le \mathbb{E}V^{(T)} \le 1$, so its one-sided boundedness constant is one, not T.

Proposition 6.2 (Truncation–Bernstein lower-tail vacuity)**.** *In the frozen family $T \ge 64$, whenever $0 < \varepsilon < 1$ and $x = \varepsilon - \beta_\kappa(T) > 0$, the proposed lower-tail probability upper bound*

$$D_{s,\kappa}(\varepsilon, T) = \exp\left[-\frac{sx^2}{2(\nu_\kappa(T) + x/3)}\right] \tag{6.9}$$

satisfies

$$\boxed{D_{s,\kappa}(\varepsilon, T) > e^{-s/160} \ge e^{-0.2} \approx 0.81873 \quad (s \le 32).} \tag{6.10}$$

Proof. Since $T^2/4 \geq 1024$ and $0 \leq \max(0, 1-\beta) \leq 1$, the minimum defining ν lies in $[80, 81]$. Also $0 < x < 1$. The exponent's positive magnitude is strictly below $s/160$, because its numerator is below s and its denominator exceeds 160. Exponentiating gives the strict floor. For $s \leq 32$, $e^{-s/160} \geq e^{-0.2}$. If $\beta \geq \varepsilon$, the cap provides no admissible positive deviation and cannot rescue the bound. Even the most permissive declared allocation is $0.10/2 = 0.05$, far below the floor.

□

This is a floor on the proposed probability upper bound, not on the actual failure probability. Thus this truncation family is vacuous uniformly over the stated parameter grid. The conclusion neither excludes a better small-ball argument nor permits the canonical h_2 term to replace control of the full sample variance.

The constant 81 specifies the two bound families just analyzed. Appendix A also establishes the sharper quadratic-chaos factor 15. Replacing 81 by 15 in the joint Chebyshev calculation gives $\sqrt{2[14 + 2/31]/1.6} \approx 4.192928 > 1$ at $s = 32$. It therefore remains vacuous on the same smaller-sample grid. This calculation does not extend the 81-based truncation proposition to every sharper concentration construction.

6.2 Paired differences and structural prior information

For disjoint pairs of common probes, write

$$D_j = \frac{(X_{a,2j-1} - X_{a,2j})^2}{2\ell_a} - \frac{(X_{0,2j-1} - X_{0,2j})^2}{2\ell_0}. \tag{6.11}$$

These variables are conditionally iid across pairs, with $\mathbb{E}D_j = \Delta$, while retaining dependence between actions within each pair. Their centered versions have degree at most four in the signs. The general hypercontractive fourth-moment factor is therefore 6561, not the degree-two factors 81 or 15. The audited data-only scale constructions remain budget-vacuous for $s \leq 32$ [1].

Theorem 6.3 (A valid norm prior supplies a finite paired radius)**.** *Condition on the pre-certification information $\mathcal{F}$, which fixes both actions, their positive denominators, and a finite bound $0 \leq M_0 < \infty$ satisfying $\|H_0\|_F \leq M_0$. The bases are nested: $R_a R_0 = R_a$. There are integer $n \geq 1$ disjoint pairs of fresh common Rademacher probes, conditionally iid and independent of $\mathcal{F}$. All probabilities and moments below are conditional on $\mathcal{F}$. Use the quadratic-chaos fourth-moment bound 81.*

For $\bar{D} = n^{-1}\sum_j D_j$, define

$$\bar{v} = 164 M_0^4 \left(\frac{1}{\ell_a} + \frac{1}{\ell_0}\right)^2. \tag{6.12}$$

Then for $0 < \delta < 1$,

$$\Pr\left(\Delta \leq \bar{D} + \sqrt{\frac{\bar{v}}{n}\frac{1-\delta}{\delta}}\right) \geq 1 - \delta. \tag{6.13}$$

Proof. Nesting gives $H_a = R_a H_0 R_a$, hence $\|H_a\|_F \leq M_0$ and $\sigma_x^2 \leq 2M_0^2$ for both actions. For $W = (X_1 - X_2)^2/2$, centered iid expansion yields

$$\mathrm{Var}(W) = \frac{\mu_4 + \sigma^4}{2} \leq 41\sigma^4. \tag{6.14}$$

Indeed $\mathbb{E}(Z_1 - Z_2)^4 = 2\mu_4 + 6\sigma^4$; divide by four and subtract σ^4. The triangle inequality for centered L^2 norms now gives

$$\mathrm{Var}(D_j) \le 41\left(\frac{\sigma_a^2}{\ell_a} + \frac{\sigma_0^2}{\ell_0}\right)^2 \le \bar{v}. \tag{6.15}$$

Independence across pairs gives $\mathrm{Var}(\bar{D}) \le \bar{v}/n$. Cantelli's one-sided inequality yields the displayed radius. To see its constant directly, for a centered variable Y of variance v, Markov's inequality on $(Y+b)^2$ gives $\Pr(Y \ge t) \le (v+b^2)/(t+b)^2$. Minimizing over $b \ge 0$ at $b = v/t$ gives $v/(v+t^2)$. Apply this to $Y = \Delta - \bar{D}$ and solve for failure probability δ.

□

If $M_0 = 0$, both residuals vanish and the radius-zero conclusion is deterministic. This avoids division by zero in the auxiliary Cantelli calculation. If the norm prior is only probabilistically valid, its failure probability enters the total confidence budget. A finite radius alone does not establish useful acceptance, affordable prior estimation, or safe fallback relative to the original baseline.

Supporting moment calculations. Appendix A gives the corrected quadratic-chaos fourth moment and the exact scale-estimation boundary. These refinements do not turn a degree-two bound into a degree-four one.

7 Experimental validation

The following findings concern the recorded frozen experiments, not universal optimality or safety claims. All detailed protocols and archived outputs are mapped in the accompanying source package [1].

7.1 Why the bridge preferred one extra column

The frozen rank-deficient bridge used $d = 500$, $k \in \{5, 15, 30\}$, $\eta \in \{0, 10^{-14}, 10^{-10}, 10^{-6}\}$, budgets $m \in \{80, 160, 240\}$, and 200 paths per setup. Matrix-orientation seeds were 52000–52002, one per rank and reused across tail levels; basis seeds were 70000–70199. Range sketches were nested coordinate-Rademacher prefixes. The frozen rank-aware QR used relative tolerance 10^{-12} and absolute tolerance zero, relative to its sampled-column reference scale—not a universal eigenvalue threshold.

The allocation grids ended at 36, 76, and 116 for the respective budgets, retaining a residual floor of eight in the full-rank planning grid. There were 554,400 trial-allocation rows. At $\eta = 10^{-6}$, **all 138,600 individual rows** satisfied $r_q = q$, not merely an average-rank statement.

Here $q_X^\star$ denotes the chosen minimizer of the finite empirical mean risk curve on the tested allocation grid, using the frozen plateau/tie convention; it is not an individual path's minimizer or the unknown population optimum. In all nine rank–budget combinations at that tail level, the recorded minima were

$$q_{\mathrm{rank}}^\star = k, \qquad q_G^\star = q_R^\star = k + 1. \tag{7.1}$$

At $m = 160$, the reproduced means were:

Table 2: Mean conditional risks at the knee (budget 160)

k	Gaussian risk at k	Gaussian risk at $k+1$	Rademacher risk at k	Rademacher risk at $k+1$
5	6.5616×10^{-11}	6.6759×10^{-12}	5.8695×10^{-11}	7.9857×10^{-14}
15	1.2123×10^{-8}	7.5641×10^{-12}	1.2059×10^{-8}	2.4140×10^{-13}
30	1.5521×10^{-11}	9.5731×10^{-12}	6.5597×10^{-12}	5.9134×10^{-13}

The denominator thresholds $(D-2)/D$ are respectively 148/150, 128/130, and 98/100. They require energy reductions of only 1.3333%, 1.5385%, and 2%. To compare means correctly, multiply each mean-risk ratio by $(D-2)/D$ to obtain the corresponding ratio of mean energies. The large reductions in the table are not explained by the denominator; the denominator penalizes the extra direction.

The mechanism audit [1] connected poorly conditioned near-square signal blocks, principal-angle errors, and tail-dominated conditional risk along nested paths. However, median behavior often preferred k, Gaussian bootstrap intervals for the mean difference at $m = 160$ included zero, and sensitivity experiments produced shifts of zero, one, two, or more. Rare one- or two-path failures can dominate a finite mean. The strongest conclusion is therefore **modest oversampling can protect against rare capture failures; exactly one extra column is not a universal optimum**.

7.2 A sharp pilot knee missed an entire dominant direction

The recovered orientation experiment used $d = 100, m = 60$, six spectra, 30 Haar orientations plus a coordinate orientation, and ten paths per orientation. The heuristic first built an eight-column Rademacher pilot and cached its basis products. Among positive Ritz values exceeding $10^{-12}\theta_1$, it required at least four values and selected the largest adjacent logarithmic gap. If that gap was at least 1.5, it used $q = \min(q_{\max}, \max(8, j_{\text{gap}} + 2))$; otherwise it used $q_0 = \min(q_{\max}, \lfloor m/3 \rfloor)$, where $q_{\max} = \min(d, \lfloor (m-2)/2 \rfloor)$. Pilot products were reused, and fresh residual probes were Rademacher. No contrast filter was active.

For the rank-five step spectrum with $\eta = 0.001$, every gate triggered and selected $(q, r, \ell) = (8, 8, 44)$. Standard Hutch++ used $(20, 20, 20)$. The finite observed comparison was:

Table 3: Recovered rank-five step experiment

Orientation group	Paths	Gated MSE	Standard MSE	Gated / Standard
Haar orientations	300	2.839433×10^{-7}	1.672813×10^{-6}	0.169740
Coordinate aligned	10	3.303942×10^{-3}	7.209998×10^{-7}	4582.446

The coordinate result is not erased by the rotated gain. Coordinate-Rademacher sketches are not rotation invariant, and the exact Rademacher residual risk depends on coordinates too.

In the seed-2026 replay, rows three and four of the failing signal block are exact negatives. Rational elimination verifies rank four. Therefore $w = (e_3 + e_4)/\sqrt{2}$ satisfies $S^\top w = 0$ and $Aw = w$. Proposition 4.3 applies. Numerically, $\|Q^\top w\|_2 \approx 8.07 \times 10^{-14}$ and the largest signal-subspace angle

is 90 degrees. The small tail supplies additional accepted directions, so full numerical rank eight does not imply five captured signal directions.

The observed Ritz matrix has four values near one and four near 0.001. Its gap is approximately 6.907721 and contrast approximately 69,077. A contrast threshold of five would not exclude it. The pilot observes an excellent but incomplete knee. The alternative $A' = A - (1-\eta)ww^\top$ from Proposition 4.3 has only four dominant eigenvalues and exactly the same AS and AQ observations.

The failing basis has $E_G \approx 1.000091$, $E_R \approx 0.500004$, and exact conditional Rademacher risk **0.02272744518**. Ordinary gated coordinate paths are near 1.31×10^{-7}. The failing path's observed squared error is 0.03303752, just 1.45364 times its own conditional risk. It contributes **99.9948% of total conditional risk** and **99.9942% of observed squared error** among the ten coordinate trials. Thus the large error is already encoded in the basis; it is not explained solely by an unlucky final probe.

The missed projector illustrates coordinate dependence: $g^\top ww^\top g = 1 + g_3 g_4$, taking values zero and two. This illustration is not used to lower-bound the variance of a sum without checking covariance; the total energies above were computed separately. Plain Rademacher Hutchinson is exact on the original diagonal matrix, but a randomized residual projector can create off-diagonal variance.

This event has **positive oversampling** $q - k = 3$ and exact discrete signal-rank failure. It must not be conflated with the invertible-but-poorly-conditioned zero-oversampling mechanism. Both invalidate "correct numerical dimension implies adequate capture," for different reasons.

The focused coordinate audit [1] replayed 310 first-spectrum paths and recorded 620 Standard/gated basis diagnostics. Diagnostic work used 74,400 replay queries plus 2,000 independent energy-check queries, separately accounted from each 60-query estimator. The legacy `Gaussian_Hutch_pplus` comparator has a Gaussian range sketch but Rademacher final probes, and here uses $q = 15$ rather than Standard's 20. It is not an all-Gaussian rotation-invariant control.

7.3 Fresh probes contain a useful signal, not a theorem

The offline risk-discrimination experiment reconstructed 200 frozen paths for each of three ranks and two tail levels. Each path received four batches of 50 certification repetitions with nested $s \in \{4, 8, 16, 32\}$. The same fresh probes served all actions, budgets, estimators, and both tail levels, so these comparisons are paired.

The estimators were ordinary sample variance, the independent-pair mean, and two medians of block means. For $W_j = (X_{2j-1} - X_{2j})^2/2$, $\mathbb{E}W_j = \sigma^2$; the sample variance and pair mean are unbiased families. `mom_w1` takes the median of individual W_j values; `mom_w2` takes the median of two-W block means. At $s = 16$ they have eight and four blocks. Neither is generally unbiased. At $s = 4$ both coincide with the pair mean, and at $s = 8$ `mom_w2` does too; these degeneracies are not independent evidence.

The primary comparison was $q = k + 1$ against $q = k$ at $m = 160$, $\eta = 10^{-6}$, $s = 16$. The empirical guard used $\varepsilon = 1/3$, so acceptance required an estimated candidate risk no more than half the estimated baseline risk. Rejection used the reverse ratio; equality or insufficient separation meant abstention. No finite-sample confidence event was attached to this rule.

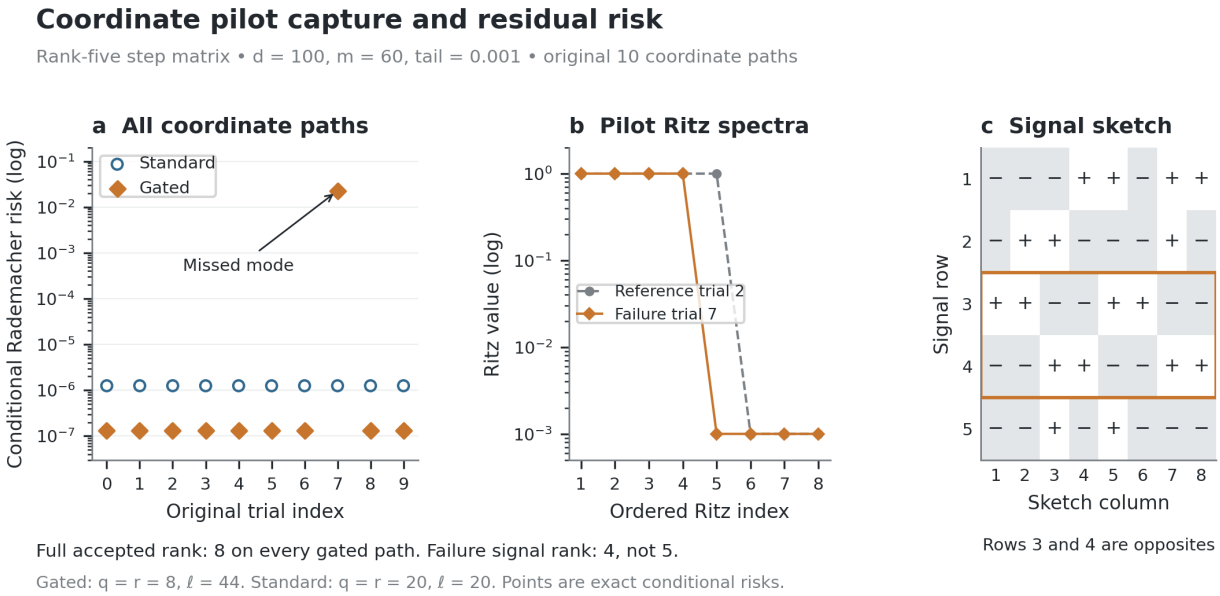


Figure 3: **Full rank with incomplete capture.** (a) Exact conditional Rademacher risks for all ten coordinate paths at $d = 100$, $m = 60$, $k = 5$, and $\eta = 0.001$. These are not observed squared errors. (b) Pilot Ritz values for the highest-risk path (trial 7) and the median-risk member of the other nine paths (trial 2). (c) The failing 5×8 signal sketch has opposite third and fourth rows. Its signal rank is four although the accepted basis rank is eight. The selected spectra illustrate the failure; panel (a) shows the full coordinate sample.

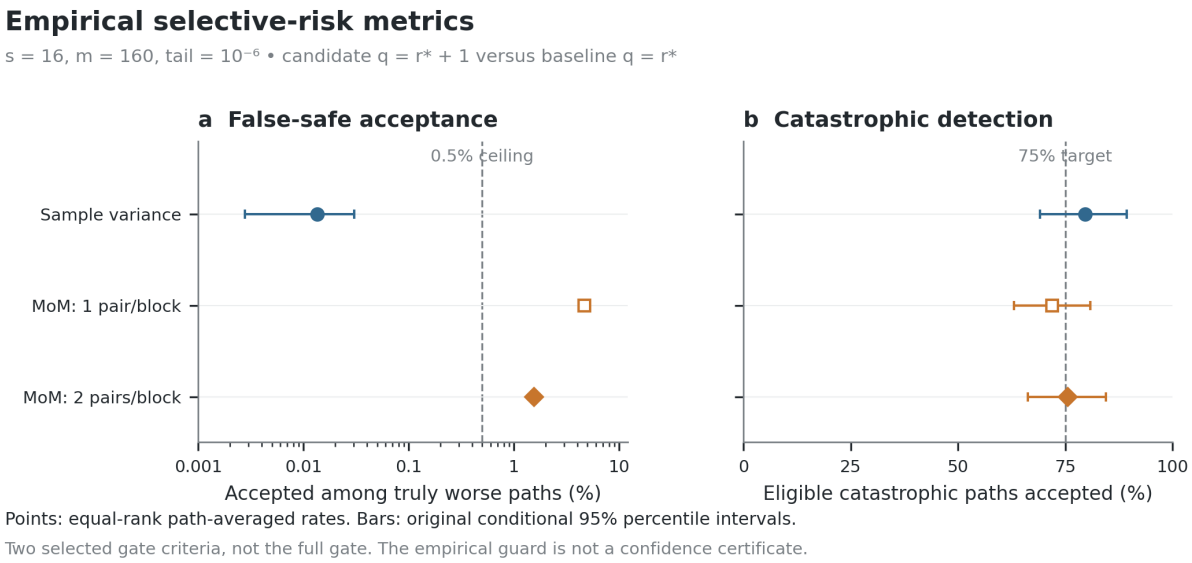


Figure 4: **Empirical direct-risk discrimination.** At $m = 160$, $\eta = 10^{-6}$, and $s = 16$, compare $q = k + 1$ with $q = k$ using the factor-two empirical guard. (a) Acceptance among truly worse paths. (b) Acceptance among true-better paths in the within-rank top 5% of baseline risk. There are 200 paths per rank and 200 repetitions per path. Points average repetitions, eligible paths, then ranks equally. Bars are 95% percentile intervals from 10,000 eligible-path, rank-stratified bootstrap replicates. Dashed lines show the selected descriptive thresholds, not confidence guarantees.

Table 4: Primary sample-variance operating rates (16 certification probes)

Sample-variance primary metric	Point estimate	Conditional 95% bootstrap interval
False-safe rate among truly worse paths	0.0134%	[0.0028%, 0.0305%]
False-rejection rate among truly better paths	0.5157%	[0.2991%, 0.7722%]
Top-5% true-better catastrophic detection	79.6667%	[69.0662%, 89.2833%]
Acceptance among all truly better paths	52.0991%	[40.3055%, 63.5069%]

Rates average repetitions within each frozen path, eligible paths within rank, then ranks equally. False-safe denominators contain only truly worse paths; false-rejection denominators only truly better paths. Catastrophic sets are defined within rank from the exact baseline risk and restricted to truly better primary candidates. The 10,000-replicate bootstrap resamples eligible path clusters within rank, not individual repetitions. Empty required rank populations are unevaluable rather than silently discarded. Truth ties and undefined zero-risk relative errors are treated separately.

The benign $\eta = 10^{-10}$ control false-safe estimate was 0.0008%, with interval [0%, 0.0025%]. The two MoM false-safe estimates were 4.6971% and 1.5478%, compared with 0.0134% for sample variance. The independent-pair mean served as an ablation. These are empirical operating rates, not confidence guarantees for accepted actions.

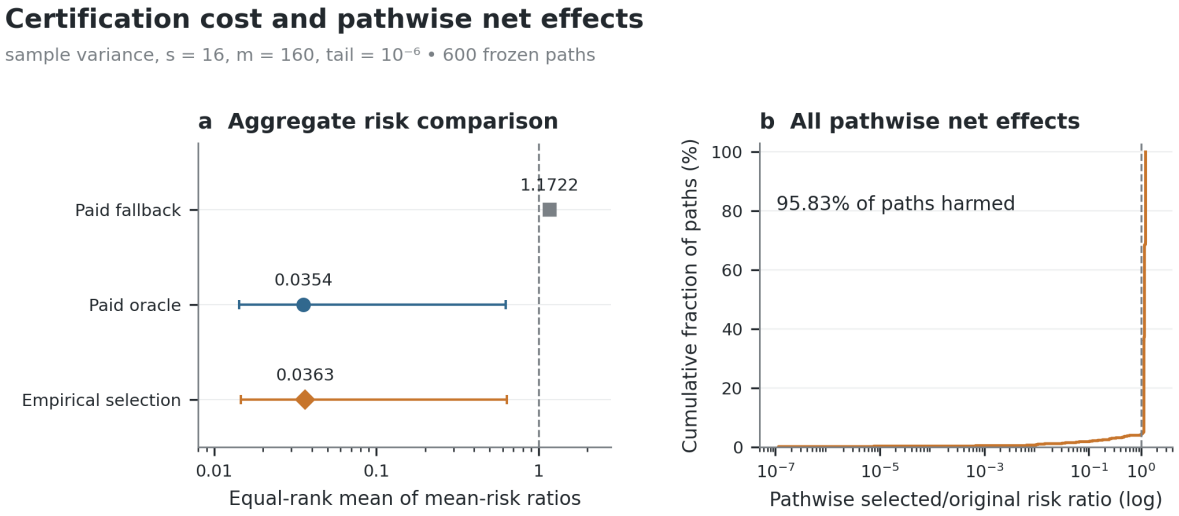


Figure 5: **Certification cost and rare-path protection.** The same 600 paths use sample variance with $s = 16$, $m = 160$, and $\eta = 10^{-6}$. The reference is the unstarted adjacent baseline. (a) Equal-rank averages of within-rank mean-risk ratios, with 95% percentile intervals from 10,000 shared rank-stratified path-bootstrap replicates. The paid oracle uses exact offline risk; paid fallback always returns to the baseline basis after costs. (b) All 600 pathwise selection/original risk ratios, with no confidence band. Protecting rare costly baselines improves the aggregate mean despite increasing risk on most paths.

7.4 Paying for certification changed the conclusion

The budget-charged emulation deducted construction and certification costs before final estimation. Across 600 frozen paths, the equal-rank average of within-rank **ratios of mean risks** was 0.036329, with conditional bootstrap interval [0.014628, 0.639830]. The paid oracle was 0.035411 and the paid fallback 1.172197.

Nevertheless, **575 of 600 individual paths—95.83%—were harmed** relative to their original baseline. The pooled median pathwise ratio was 1.160714. The historical value 1.172197 described an equal-rank average of within-rank medians in the relevant summary, not that pooled median. Means of ratios, ratios of means, and averages of medians are distinct statistics.

7.5 Common-probe cancellation was strong on average, weaker on catastrophes

The common-probe comparison gave an equal-rank empirical paired/independent variance ratio of **0.151032**, with interval **[0.137069, 0.166735]**, approximately 84.9% lower. The independent benchmark is the plug-in sum of marginal variances. Catastrophic paths had mean pairing ratio **0.899565**, compared with **0.111635** for ordinary paths. Thus the strongest cancellation did not occur on the rare paths most relevant to aggregate risk [1].

7.6 Additional gains remain exploratory

The matrix-free Hessian benchmark used a 4,254-parameter convolutional network on synthetic images. It used the same eight-column pilot rule with logarithmic-gap threshold 1.2, oversampling parameter two, and no contrast filter. The model has ten outputs, but its labels are only zero and one. The Hessian need not be PSD; Theorems 2.2–2.3 apply to a symmetric operator, whereas PSD spectral interpretations need additional justification. The floating-point reference trace was approximately 31.581270, obtained using 4,254 reference Hessian–vector products outside the estimator budgets.

Table 5: Exploratory synthetic-data Hessian results

Query budget	Standard Hutch++ MSE	Gated MSE
30	5.858498	5.334005
60	1.417330	0.841176
90	1.006494	0.570674
120	0.457086	0.370210

There were 30 trials per method/budget and 480 total across four methods. These point estimates show operator feasibility and gains on this model, not broad ML generalization, a certified safety guarantee, or a wall-clock advantage. The gain cannot be attributed solely to more residual samples because the basis changes too. Sources: Hessian benchmark [1] and recovery audit [1].

The gate-feature development study [1] contains 23 configurations. Its thresholds were assessed on that same grid; fixed-q empirical minima are noisy grid references, not exact conditional-risk oracles. The later contrast filter was not active in the recovered Haar/Hessian protocols, so those results do not validate that filter.

Earlier YearPredictionMSD and Wiki-Vote studies, including quoted effective ranks, crossover budgets, and reduction factors, remain historical. Those numerical claims have not received source-level reproduction in this consolidation and are not central evidence here. A final external-data claim needs its operator definition, probe law, accounting, replication unit, and uncertainty verified. TurboQuant and leverage-score work remain supporting RandNLA preparation rather than competing main UROP narratives.

8 Discussion and conclusion

Main conclusion. Under a fixed query budget, an additional subspace direction is worthwhile only when its realized residual-risk reduction exceeds the construction and evaluation cost. Visible spectral features alone do not certify that this condition holds. The project therefore changes the emphasis from predicting a favorable split from the spectrum to justifying a decision using the residual risk that the realized estimator actually incurs.

The analysis identifies several distinctions that an adaptive trace-estimation policy must respect. Attempted sketch width is not accepted rank. Accepted rank is not dominant-subspace capture. An ideal spectral tail does not in general equal realized probe-specific residual energy. A useful empirical comparison is not a finite-sample certificate, and safe acceptance is not the same as a safe complete policy after information costs have been paid.

Capture limitations. The capture results explain two different failures. An invertible square signal sketch can be poorly conditioned, amplifying a small nonzero tail. A discrete signal sketch can also miss a direction exactly, even with positive oversampling and a full-rank accepted basis. The first mechanism helps explain tail-sensitive empirical mean curves; the second supplies an exact local pilot-transcript limitation. Neither implies a universal frequency of failure or a fixed oversampling amount that is always optimal.

Certification limitations. The certification results also require two levels of interpretation. Ordinary sample variance was useful under the preregistered empirical decision gate. Charging for certification revealed a tradeoff between aggregate mean-risk protection and harm on many ordinary paths. The explicit confidence routes considered here were too conservative at the tested sample sizes. This gap between empirical signal and available guarantees is a research question, not a license to interpret an empirical guard as a theorem.

Limitations. The conditional-risk bridges use finite frozen orientations and path populations. Their bootstrap intervals do not quantify unseen rare-event probabilities or orientation-universal performance. The Hessian and pilot-gate experiments are exploratory. The structural-prior certificate assumes valid prior information and does not establish that this information can be obtained within the same budget. Our route-specific negative results do not exclude other confidence constructions.

Future work and open problems. A focused continuation is to derive an observable, budget-feasible one-sided upper confidence bound for the common-probe difference

$$\Delta_R = \frac{\sigma_a^2}{\ell_a} - \frac{\sigma_0^2}{\ell_0}.$$

Such a result should exploit paired information when justified, charge all committed construction and certification queries, preserve the freshness of final residual probes, and specify a feasible fallback. A fresh-query capture test or a genuine small-ball analysis is an alternative direction. Another direction is structured or leverage-guided sketching, provided that score estimation and missed-direction risks are charged to the same query budget. These directions remain open problems: no new allocator is proposed or implemented in this manuscript.

The practical goal of these extensions is not merely to find a favorable allocation, but to make the allocation decision observable, budget-feasible, and defensible under the realized risk of the estimator.

References

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A Supporting moment and scale calculations

Lemma A.1 (A corrected degree-two fourth moment)**.** *C is real symmetric, nonzero, zero diagonal; $Z = g^\top Cg$ for coordinate Rademacher g. Put $\sigma^2 = 2\|C\|_F^2$ and $\kappa = \|C\|_2/\|C\|_F$.*

The corrected identity and its consequence are

$$\mathbb{E}Z^4 = 3\sigma^4 + 48\,\mathrm{tr}(C^4) - 96\sum_i [(C^2)_{ii}]^2 + 32\sum_{i,j} C_{ij}^4, \tag{A.1}$$

$$\boxed{\mathbb{E}Z^4 \le (3 + 12\kappa^2)\sigma^4 \le 15\sigma^4.} \tag{A.2}$$

Proof. Write $Z = 2\sum_{i<j} C_{ij} g_i g_j$. In its fourth-power expansion only edge multisets with even degree at every vertex survive. They consist of one edge repeated four times, two distinct edges each repeated twice, or four distinct edges forming a four-cycle. Their multiplicities in the edge expansion are respectively 1, 6, and 24. Collecting these terms into matrix traces gives the displayed identity; the corrected coefficients 48, -96, and 32 retain the repeated-edge adjustments.

For more explicit bookkeeping, let $a = \sum_{i<j} C_{ij}^4$, let b sum $C_e^2 C_f^2$ over unordered distinct edge pairs, let b_{adj} restrict to pairs sharing a vertex, and let c sum products over undirected four-cycles. The edge expansion is $16a + 96b + 384c$. Meanwhile $\sigma^4 = 16a + 32b$, $\mathrm{tr}(C^4) = 2a + 4b_{\mathrm{adj}} + 8c$, $\sum_i [(C^2)_{ii}]^2 = 2a + 2b_{\mathrm{adj}}$, and $\sum_{ij} C_{ij}^4 = 2a$. Substitution verifies every coefficient.

Now $\sum_{ij} C_{ij}^4 \le \sum_i (\sum_j C_{ij}^2)^2 = \sum_i [(C^2)_{ii}]^2$, so the combined correction after the trace term is nonpositive. Also $\mathrm{tr}(C^4) \le \|C\|_2^2 \|C\|_F^2$. Divide by $\sigma^4 = 4\|C\|_F^4$ to obtain $3 + 12\kappa^2$, and use $\kappa \le 1$. □

The same identity gives a lower bound $3 - 24\kappa^2$ on the normalized fourth moment, since $\sum_i [(C^2)_{ii}]^2 \le \|C\|_2^2 \|C\|_F^2$. Thus small κ squeezes this quadratic-chaos kurtosis toward three. It does not make three a universal lower bound, and it says nothing directly about the kurtosis of the signed degree-four $D_j - \Delta$.

Proposition A.2 (The exact scale-estimation boundary)**.** *A scalar observation has variance $v > 0$ and fourth central moment at most Kv^2, with $K \ge 1$. Estimate its variance with $n \ge 2$ iid observations and use the Lemma 5.1–Chebyshev relative radius at failure probability $0 < \delta_{\mathrm{scale}} < 1$.*

That radius is below one exactly when

$$\boxed{\delta_{\mathrm{scale}} n(n-1) > (K-1)(n-1) + 2.} \tag{A.3}$$

At $K = 3$ and $\delta_{\mathrm{scale}} = 0.05$, the first feasible integer is $n = 42$. If each observation uses two probes, this requires $s = 84$, not 82.

Proof. Lemma 5.1 bounds the relative sample-variance variance by $[(K-1) + 2/(n-1)]/n$. Chebyshev's relative radius squared divides this by δ_{scale}. Requiring it below one and multiplying by the positive $n(n-1)\delta_{\mathrm{scale}}$ gives the inequality. At $n = 41$, both sides equal 82; strict feasibility fails. At 42, the left side is 86.1 and the right side 84. The expression decreases with n, proving first feasibility at 42. □

This boundary neither proves that a norm prior is necessary for every certificate nor licenses assuming $K = 3$ for an arbitrary signed-pair observation. It diagnoses one explicit scale-estimation

route. Sources for Theorem 6.3–Proposition A.2: structural proof note [1] and the recovery correction [1].

B Reproducibility and provenance

The accompanying source package maps each theorem and empirical finding to the detailed project records [1]. It includes original schematics, numerical figure sources, and a separate development history. The three quantitative plots use the validated records; only internal study labels were removed from two plot subtitles. No plotted value or interval was changed.

Some experimental outputs required protocol-based recovery after smoke tests overwrote untracked files. The recovered orientation and Hessian results reproduce the recorded protocols, but unavailable original hashes prevent a claim of byte-identical restoration. The companion recovery audit documents this limitation.

Uncertainty follows the replication unit. Certification repetitions are clustered within paths, and paths can share orientations. The reported bootstrap intervals are conditional on the frozen population, not universal performance guarantees. The exploratory orientation and Hessian tables do not receive new confidence intervals in this manuscript.

The two conceptual figures are original, programmatically drawn schematics, not experimental measurements. AI assistance supported editorial revision, illustration code, and proof auditing. Scientific claims and authorship metadata require author verification before submission.